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PORTABLE CONTAINER SYSTEM

Abstract

A container system comprises a container that includes a container body with a storage cavity and a lid assembly releasably connected to the container. The lid assembly has a lid that includes a gasket received in a channel on the lid, and a latch pivotally connected to the lid. The container may also include a perimeter lip extending outward around an entire perimeter of the container opening of the container. The latch may include a latch rib, such that when the latch is engaged with the perimeter lip of the container, the latch rib is located over the channel of the lid. The channel on the lid that receives the gasket may also include a channel rib.

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Background/Summary

FIELD OF THE INVENTION

[0001] This disclosure relates to portable containers. More specifically, aspects of this disclosure relate to a portable container with a lid, where when the container is in a closed configuration, the container provides a waterproof and airtight seal.

BACKGROUND

[0002] Portable containers are commonly used to store various items such as food or liquids. The containers may also provide a degree of protection from incidental impacts (e.g., drops) as well as from incursion from liquids or dirt. By sealing the container, the contents are prevented from leaking onto other items when stored and also protected from incursion into the contents. Containers may be composed of rigid materials and have an opening that may be sealed with a lid.

BRIEF SUMMARY

[0003] The following presents a simplified summary of various aspects described herein. This summary is not an extensive overview, and is not intended to identify key or critical elements or to delineate the scope of the claims. The following summary merely presents some concepts in a simplified form as an introductory prelude to the more detailed description provided below.

[0004] This disclosure may relate to a portable container that includes a container system, comprising: (a) a container including: (1) a first wall; (2) a second wall opposite the first wall; (3) a third wall extending between the first wall and the second wall; (4) a fourth wall opposite the third wall extending between the first wall and the second wall; and (5) a bottom wall connected to the first wall, the second wall, the third wall, and the fourth wall at a bottom end of the container forming a storage cavity with a container opening opposite the bottom wall; and (b) a lid assembly releasably connected to the container, where the lid assembly covers the container opening when connected to the container, and where the lid assembly comprises: (1) a lid that includes a top surface and a channel on a bottom side of the lid; (2) a gasket received in the channel; and (3) a latch pivotally connected to the lid. The container may also include a perimeter lip extending outward around a portion of a perimeter of the container opening, where the perimeter lip includes a lip protrusion extending from a lower surface of the perimeter lip. The latch may include a latch rib, such that when the latch is engaged with the perimeter lip of the container, the latch rib is located over the channel of the lid. The channel may also include an upper surface that has a channel rib that extends away from the upper surface toward the bottom side of the lid, where the latch rib and the channel rib are substantially aligned when the latch is engaged with the perimeter lip of the container. The container may also include a container rib that extends upward and away from an upper surface of the perimeter lip. In some examples, the container rib may be substantially aligned with the channel rib and the latch rib. The channel may form a continuous loop, and the channel rib may also form a continuous loop. The container rib and the channel rib may contact the gasket on opposite sides of the gasket. In addition, the latch rib may contact an outward facing surface of the lid opposite the channel. The lid may comprise a receiver that includes a first axle that extends from a first side surface of the receiver and a second axle that extends from a second side surface of the receiver. The first axle may include a cylindrical portion and an end cap at a free end, where an axle clip of the latch connects to the cylindrical portion, and where the end cap has a maximum width that is larger than a diameter of the cylindrical portion. The latch may include a top wall connected to a side wall, where the side wall is substantially perpendicular to the top wall, and the side wall has a flange that engages the perimeter lip. The latch rib may extend outward from an inner surface of the top wall. Additionally, the latch may have a latch length extending from a first end of the top wall to a second end of the top wall, and the latch rib may have a latch rib length, where the latch rib length is within a range of 85 percent to 98 percent of the latch length.

[0005] Other aspects of this disclosure may relate to a lid assembly for a covering an opening of a container, the lid assembly comprising: (a) a lid that includes a top surface and a channel on a bottom side of the lid, where the channel includes an upper surface that has a channel rib that extends away from the upper surface toward the bottom side of the lid; (b) a gasket received in the channel; and (c) a latch pivotally connected to the lid, where the latch is configured to rotate between a locked configuration and an unlocked configuration. The latch may include a latch rib, such that when the latch is in the locked configuration the latch rib contacts an outward facing surface of the lid opposite the channel, and when the latch is in the unlocked configuration, the latch rib is free of contact with the outward facing surface of the lid opposite the channel. The latch rib and the channel rib may be substantially aligned when the latch is in the locked configuration. The latch may also include a top wall connected to a side wall, where the side wall is substantially perpendicular to the top wall. The side wall may include a flange that is configured to engages a perimeter lip of the container when the latch is in the locked configuration. The latch rib may be configured to apply a force to the gasket through the channel rib when the latch is moved to the locked configuration.

[0006] Still other aspects of this disclosure may relate to a container system, comprising: (a) a container that includes (1) a container body including a storage cavity that is accessed through a container opening, (2) a perimeter lip extending laterally outward around at least a portion of a perimeter of the container opening, where the perimeter lip includes: a lip protrusion extending from a lower surface of the perimeter lip; and a container rib that extends upward and away from an upper surface of the perimeter lip; and (b) a lid assembly releasably connected to the container, where the lid assembly covers the container opening when connected to the container opening. The lid assembly may comprise: (1) a lid that includes a top surface; a channel located on a bottom side of the lid, where the channel includes an upper surface that has a channel rib that extends away from the upper surface toward the bottom side of the lid; and a plurality of receivers positioned laterally outward of the top surface, where a first receiver of the plurality of receivers may include a first side surface and a second side surface, and where a first axle extends from the first side surface and a second axle extends from the second side surface; (2) a gasket received in the channel; and (3) a plurality of latches, where a first latch of the plurality of latches includes a top wall, a side wall connected to the top wall, and a pair of axle clips extending from an inner surface of the top wall, wherein the side wall includes a flange that engages a bottom surface of the lip protrusion of the perimeter lip when the container system is in a locked configuration to compress the gasket between the container rib and the channel rib. The first axle may include a cylindrical portion and an end cap at a free end, where an axle clip of the pair of axle clips connects to the cylindrical portion. The first latch may further include a curved wall connected to the top wall opposite the side wall, where the curved wall has an outer surface that includes a restraint wall such that when the first latch is rotated upward to an unlocked configuration, the restraint wall may contact a protrusion extending upward from an upward facing surface of the first receiver. The first latch may further include a latch rib that extends from the inner surface of the top wall, such that when the first latch is in the locked configuration, the latch rib contacts an outward facing surface of the lid opposite the channel. The channel may be located laterally outward of the first axle and the second axle of each receiver.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0007] FIG. **1** depicts a top, right front perspective view of the portable container system in a locked configuration according to aspects disclosed herein;

[0008] FIG. **2** depicts a bottom, right front perspective view of the portable container system of FIG. **1** according to aspects disclosed herein;

[0009] FIG. **3** depicts a top view of the portable container system of FIG. **1** according to aspects

disclosed herein;
[0010] FIG. **4** depicts a front view of the portable container system of FIG. **1** according to aspects disclosed herein;
[0011] FIG. **5** depicts a right side view of the portable container system of FIG. **1** according to aspects disclosed herein;
[0012] FIG. **6** depicts cross-sectional view of the portable container system of FIG. **4** along line **6-6** according to aspects disclosed herein;
[0013] FIG. **7** depicts an enlarged view of a portion of the portable container system of FIG. **6** according to aspects disclosed herein;
[0014] FIG. **8** depicts a top, rear perspective view of the portable container system of FIG. **1** in an unlocked configuration according to aspects disclosed herein;
[0015] FIG. **9** front view of the portable container system of FIG. **8** according to aspects disclosed herein;
[0016] FIG. **10** depicts a cross-sectional view of the portable container system of FIG. **8** along line **10-10** according to aspects disclosed herein;
[0017] FIG. **11** depicts an enlarged view of a portion of the cross-sectional view of the portable container system of FIG. **8** according to aspects disclosed herein;
[0018] FIG. **12** depicts an enlarged partial perspective view of the portable container system of FIG. **8** according to aspects disclosed herein;
[0019] FIG. **13** depicts a top, front perspective view of the lid of the portable container system of FIG. **1** according to aspects disclosed herein;
[0020] FIG. **14** depicts a bottom, front perspective view of the lid of the portable container system of FIG. **13** according to aspects disclosed herein;
[0021] FIG. **15** depicts a top view of the lid of the portable container system of FIG. **13** according to aspects disclosed herein;
[0022] FIG. **16** depicts an enlarged partial perspective view of the lid of the portable container system of FIG. **13** according to aspects disclosed herein;
[0023] FIG. **17** depicts an enlarged partial top view of the lid of the portable container system of FIG. **13** according to aspects disclosed herein;
[0024] FIG. **18** depicts a partial side cross-sectional view of the lid of the portable container system of FIG. **15** along line **18-18** according to aspects disclosed herein;
[0025] FIG. **19** depicts a top, front perspective view of a latch of the portable container system of FIG. **1** according to aspects disclosed herein;
[0026] FIG. **20** depicts a bottom, rear perspective view of the latch of FIG. **19** according to aspects disclosed herein;
[0027] FIG. **21** depicts a side view of the latch of FIG. **19** according to aspects disclosed herein;
[0028] FIG. **22** depicts a partial perspective view of the latch of FIG. **19** according to aspects disclosed herein;
[0029] FIG. **23** depicts a top, front perspective view of the container of the portable container system of FIG. **1** according to aspects disclosed herein;
[0030] FIG. **24** depicts a side view of the container of the portable container system of FIG. **23** according to aspects disclosed herein;
[0031] FIG. **25** depicts a top, front perspective view of an alternate portable container system in a locked configuration according to aspects disclosed herein; and
[0032] FIG. **26** depicts a top, front perspective view of an alternate portable container system in a locked configuration according to aspects disclosed herein.

DETAILED DESCRIPTION

[0033] In the following description of various example structures, reference is made to the accompanying drawings, which form a part hereof, and in which are shown by way of illustration various example devices, systems, and environments in which aspects of the invention may be practiced. It is to be understood that other specific arrangements of parts, example devices, systems, and environments may be utilized and structural and functional modifications may be made without departing from the scope of the present disclosure.

[0034] Also, while the terms “top,” “bottom,” “front,” “side,” “rear,” and the like may be used in this specification to describe various example features and elements of the invention, these terms are used herein as a matter of convenience, e.g., based on the example orientations shown in the figures or the orientation during typical use. Additionally, the term “plurality,” as used herein, indicates any number greater than one, either disjunctively or conjunctively, as necessary, up to an infinite number.

[0035] The terms “connect,” or “attach” as used herein indicates that components, surfaces, or features and the like may be directly or indirectly (i.e. through an intermediary) joined, linked or attached. In addition, “pivotally connected” as used herein, indicates that the components or features are directly or indirectly coupled together such that the components can rotate relative to each other while still being coupled together either directly or indirectly. Examples of a “pivotally connected” joint may include a pin inserted into an opening arranged in each of the components to “pivotally connect” the components.

[0036] As used herein, the term “substantially” means mostly, or almost the same as, within the constraints of sensible commercial engineering objectives, costs, manufacturing tolerances, and capabilities in the field of manufacturing the article being formed. For example, the term, “substantially aligned,” as the term is used herein, means that a first line, segment, plane, edge, surface, center plane of an element, or center axis of an element, etc. is collinear or coplanar from with another line, plane, edge, surface, center plane of an element or center axis of an element, etc. in some instances within a tolerance range of +1 mm, or within a tolerance range of +2 mm, or within a tolerance range that is appropriate for manufacturing of the described elements. Similarly, the term “substantially coplanar” as used herein, indicates that the elements of any two or more surfaces or edges may fall within a tolerance zone of parallel planes in some instances within a tolerance range of +1 mm, or within a tolerance range of +2 mm, or within a tolerance range that is appropriate for manufacturing of the described elements.

[0037] The term “approximately” means close to, or about, a particular value, within the constraints of sensible commercial engineering objectives, costs, manufacturing tolerances, and capabilities in the field of manufacturing the article being formed.

[0038] “Substantially coplanar” as used herein, indicates that the elements of any two or more surfaces or edges fall within a tolerance zone of parallel planes within a range of +/−2 mm of each other.

[0039] “Substantially parallel,” as the term is used herein, means that a first line, segment, plane, edge, surface, etc. is approximately (in this instance, within 5%) equidistant from with another line, plane, edge, surface, etc., over at least 50% of the length of the first line, segment, or edge, or over at least 50% of the area of the plane or surface, etc. In some examples, lines, segments, or edges may be considered “substantially parallel” if one such a line, segment, or edge is approximately equidistant (65%) to another respective line, segment, or edge over at least 60%, at least 75%, at least 85%, at least 90%, or even at least 95% of a length of either of the lines, segments, or edges being considered. Additionally, planes or surfaces may be considered “substantially parallel” if one plane or surface is approximately equidistant (65%) to another respective plane or surface over at least 60%, at least 75%, at least 85%, at least 90%, or even at least 95% of a surface area of either of the planes or surfaces being considered.

[0040] “Substantially perpendicular,” as the term is used herein, means that a first line, segment, plane, edge, surface, etc. is approximately (in this instance, within 5%) orthogonal from with another line, plane, edge, surface, etc., over at least 50% of the length of the first line, segment, or edge, or over at least 50% of the area of the plane or surface, etc. In some examples, lines, segments, or edges may be considered “substantially perpendicular” if one such a line, segment, or edge is approximately orthogonal (65%) to another respective line, segment, or edge over at least 60%, at least 75%, at least 85%, at least 90%, or even at least 95% of a length of either of the lines, segments, or edges being considered. Additionally, planes or surfaces may be considered “substantially perpendicular” if one plane or surface is approximately orthogonal (65%) to another respective plane or surface over at least 60%, at least 75%, at least 85%, at least 90%, or even at least 95% of a surface area of either of the planes or surfaces being considered.

[0041] Nothing in this specification should be construed as requiring a specific three dimensional orientation of structures in order to fall within the scope of this invention. The reader is advised that the attached drawings are not necessarily drawn to scale.

[0042] Generally, this disclosure relates to a portable container system that includes a container and a lid assembly, where the container system may have a locked configuration that prevents access to the storage cavity of the container and an unlocked configuration that allows the lid assembly to be removed from the container to allow access to the storage cavity.

[0043] As shown in FIGS. **1-24**, the portable container system **100** may comprise a container **110** and a lid assembly **140** that is releasably connected with the container **110**. The container system **100** may have an unlocked configuration that allows for the removal of the lid assembly **140**, and a locked configuration that prevents the removal of the lid assembly **140**. The lid assembly **140** may also comprise a plurality of latches **170** that may be pivotally connected to the lid **142**. Each latch **170** may engage a perimeter lip **130** on the container **110** when the container system **100** is in a locked configuration, and a latch **170** may be disengaged (e.g., free of contact) from the perimeter lip **130** of the container **110** when the container system **100** is in an unlocked configuration.

[0044] The container **110** may have a container body **111** that has a storage cavity **124** that is accessed through a container opening **126**. While the container **110** in the illustrated examples has a substantially rectangular profile, the container **110** is not limited to a rectangular profile as the container **110** may have a substantially circular profile, a substantially elliptical profile, or other geometric shape. The exemplary container body **111** may have a first wall **112**, a second wall **114** opposite the first wall **112**, a third wall **116** extending between the first wall **112** and the second wall **114**, and a fourth wall **118** opposite the third wall **116** extending between the first wall **112** and the second wall **114**. A bottom wall **120** may connect the first wall **112**, the second wall **114**, the third wall **116**, and the fourth wall **118** at a bottom end **122** of the container **110** forming a storage cavity **124** with a container opening **126** opposite the bottom wall **120** at an upper end **128** of the container **110** as shown in FIG. **23**. The walls **112**, **114**, **116**, **118**, **120** may be connected with a smooth juncture (e.g., rounded corners) such that the container body **111** is a smooth continuous structure. A perimeter lip **130** may extend laterally outward at the upper end **128** of the container **110**, where the perimeter lip **130** may extend around an entire perimeter (e.g., the distance around the walls **112**, **114**, **116**, **118**) of the container opening **126** such that the perimeter lip **130** extends outward around the first wall **112**, the second wall **114**, the third wall **116**, and the fourth wall **118** and forms a continuous loop. The perimeter lip **130** may include a lip protrusion **132** extending from a lower surface **134** of the perimeter lip **130**, where the lip protrusion **132** extends downward and away from the lower surface **134** of the perimeter lip **130**. A groove **136** may be formed between the perimeter lip **130** and the lip protrusion **132**. The container **110** may include supports **138** that connect the lip protrusion **132** and its corresponding wall **112**, **114**, **116**, **118** to help strengthen the perimeter lip **130** and lip protrusion **132**. The bottom wall **120** may include a plurality of feet **125** that extend from the bottom surface **123** as shown in FIG. **2**.

[0045] The lid assembly **140** may be removably connected to the container **110**. When attached, the lid assembly **140** covers the container opening **126** of the container **110** and prevents access to the storage cavity **124**. When the lid assembly **140** is removed, access to the storage cavity **124** is allowed. The lid assembly **140** may comprise a lid **142**, a plurality of latches **170** that are pivotally connected to the lid **142** and a gasket **200**. Each latch **170** may include a top wall **172** connected to a side wall **174**, where the side wall **174** is substantially perpendicular to the top wall **172**. The side wall **174** may further include a flange **176** that may extend laterally inward and away from an inner surface **177** of the side wall **174**. The flange **176** may selectively engage the perimeter lip **130**, such when the flange **176** engages the perimeter lip **130** and/or the lip protrusion **132** of the container **110**, the latch **170** is in a locked configuration as shown in FIGS. **6** and **7**, and when the flange **176** is free of contact with the lip protrusion **132** and/or the perimeter lip **130**, the latch **170** is in an unlocked configuration as shown in FIGS. **10-11**. When in latch **170** is in a locked configuration, the flange **176** may engage a bottom surface **133** of the lip protrusion **132**. When the flange **176** engages the lip protrusion **132**, a latch rib **182** on the top wall **172** may exert a downward force onto the gasket **200** positioned within a channel **152** on a bottom side **154** of the lid **142**. In some embodiments, when the container system **100** is in the locked configuration, the lid **142** may be free of contact with the container **110**, such that only the gasket **200** and the latches **170** contact both the lid **142** and the container **110**. In some embodiments, various portions of the lid **142** may be in contact with the container **110** when the lid **142** is placed on the container **110**. For example, one or more portions of the lid **142** may contact the perimeter lip **130** to align the lid **142** with the container **110**.

[0046] FIGS. **13-18** illustrate the lid **142** of the lid assembly **140**. The lid **142** includes top surface **144**, a lid protrusion **146** extending upward from and around an outer edge of the top surface **144**. In some embodiments, the top surface **144** may be planar. The lid **142** may also include a plurality of receivers **148** that are configured to receive and secure a latch **170** such that each receiver **148** may receive an individual latch **170**. In some embodiments, each receiver **148** may be located laterally outward of the top surface **144** and the lid protrusion **146**. Additionally, each receiver **148** and its corresponding latch **170** may be centered along its respective wall **112**, **114**, **116**, **118**. In other embodiments, receivers and their corresponding latches may not be centered along their respective walls. A corner cap **150** may be formed between each receiver **148**.

[0047] As shown in FIG. **3**, when in the locked configuration, the outer surface **175** of the side wall **174** of each latch **170** may be substantially coplanar with an outer edge **151** of a corner cap **150** that is nearest to the latch **170**. In addition, an upper surface **173** of the top wall **172** of each latch **170** may not extend beyond an upper surface **147** of the lid protrusion **146** when the latch **170** is in a locked configuration. In some examples, the upper surface **173** may be substantially coplanar with the upper surface **147** of the lid protrusion **146**.

[0048] The lid **142** may also include a channel **152** on a bottom side **154** of the lid **142**, where the channel **152** receives the gasket **200**. The channel **152** may form a continuous loop which is located laterally outward of the top surface **144**. The channel **152** may include a channel opening **156** that is open to the bottom side **154** of the lid **142** and an upper surface **158** opposite the channel opening **156**. The channel **152** may be substantially U-shaped and/or may have substantially vertical side surfaces adjacent the upper surface **158**. The upper surface **158** may include a channel rib **160** that extends away from the upper surface **158** toward the channel opening **156** and the bottom side **154** of the lid **142**. The channel rib **160** may form a continuous loop within the channel **152**. Additionally, in some embodiments, the channel rib **160** may be centered on the upper surface **158**. The channel **152** may be positioned under each receiver **148** such that when a latch **170** is in a locked configuration, the top wall **172** of the latch **170** is located above the channel **152** and the gasket **200**. The channel **152** may also be located laterally outward of axles **161** positioned in each receiver **148**.

[0049] The plurality of receivers **148** may each be configured to individually secure a latch **170** of the plurality of latches **170**. Each receiver **148** may comprise a pair of axles **161**, where each axle **161** has a cylindrical portion **162** that extends inward from a side surface **164** of the receiver **148** and an end cap **166** at a free end of the axle **161**. Each side surface **164** may be on opposite sides of the receiver **148** such that the pair of axles **161** extend toward each other. The cylindrical portion **162** may define an axis of rotation for each latch **170**. The end cap **166** may have a maximum width that is larger than a maximum width and/or diameter of the cylindrical portion **162**. The end cap

166 may not be centered on a longitudinal axis of the cylindrical portion **162** and may therefore be arranged asymmetrically with respect to the longitudinal axis of the cylindrical portion **162**. In some examples, the end cap **166** may comprise a projection **167** that extends beyond the diameter of the cylindrical portion **162**.

[0050] The plurality of latches **170** may be pivotally connected to the lid **142**. Each latch **170** may include a pair of axle clips **178** that connect to the pair of axles **161** of each receiver **148**. An axle clip **178** may include a rounded internal surface that allows the latch **170** to rotate around the cylindrical portion **162** of each axle **161**. The axle clip **178** may connect to the cylindrical portion **162** of the axle **161** with a friction or snap fit. The latch **170** may also include a recess **181** adjacent to each axle clip **178**. The recess **181** or at least a portion of the recess **181** may be located on an inner surface **180** of the top wall **172**. Each recess **181** may receive a corresponding end cap **166** of each axle **161**. In some examples, the projection **167** of the end cap **166** may contact and/or frictionally engage with recess **181** to help hold each latch **170** in an unlocked configuration. For examples, the frictional interface between the recess **181** and the end cap may hold the latch **170** in any angular position with respect to the lid **142** until acted upon by a user. Additionally or alternatively, an inward facing surface **165** of the end cap **166** may frictionally engage a side surface **179** of an axle clip **178** of the latch **170** to hold each latch **170** in an unlocked configuration such that the frictional interface between the side surface **179** and the inward facing surface **165** may hold the latch **170** in any angular position with respect to the lid **142** until acted upon by a user. The end cap **166** helps to prevent the axle clip **178** from coming off the axle **161** if the axle **161** or latch **170** were to deform or bend, such as when the container system **100** is dropped. The end cap **166** may also help to center the latch **170** within the receiver **148**. In addition, the end cap **166** may also help to prevent the latch **170** from moving from a locked configuration to an unlocked configuration when the container system **100** is dropped or subject to an impact load to help improve the overall durability of the container system **100**.

[0051] The container system **100** may include multiple features to ensure the gasket **200** provides a seal between the container **110** and the lid **142** to prevent any leakage of the contents out of the storage cavity **124** or any leakage of external fluids or debris into the storage cavity **124**. The container **110** may include a container rib **139** that extends upward and away from an upper surface **135** of the perimeter lip **130**. In some examples, the container rib **139** may form a continuous loop along the perimeter lip **130**, while in other examples, it is contemplated that the container rib **139** may be intermittent. As discussed above, the channel **152** that receives the gasket **200** may include a channel rib **160** that extends downward from the upper surface **158** of the channel **152** toward the channel opening **156**. The channel rib **160** and the container rib **139** may both contact the gasket **200** on opposite sides of the gasket **200** as shown in FIG. **7**. In addition, in some embodiments the container rib **139** and the channel rib **160** may be substantially aligned to apply consistent pressure on the gasket **200**. This consistent pressure on the gasket **200** helps to ensure a tight waterproof seal formed by the compression of the gasket **200**. The gasket **200** may be compressed approximately 30 percent (e.g. measured in a cross-sectional profile) from an undeformed state or compressed within a range of 20 percent and 40 percent from an undeformed state, or in some examples, the gasket **200** may be compressed within a range of 15 percent and 40 percent from an undeformed state. The force to compress the gasket **200** may vary by container size. For example, a larger container with a larger gasket may require a larger force to sufficiently compress the gasket **200**. The gasket **200** may also include a centrally located hollow cavity **202**. Additionally, each latch **170** may include a latch rib **182** that extends away from an inner surface **180** of the top wall **172**. When the latch **170** is in the locked configuration, the latch rib **182** may be located over the channel **152** and contact and exert a force on an outward facing surface **155** opposite the upper surface **158** of the channel **152**. To further focus the pressure on the gasket **200**, when the latch **170** is in the locked configuration, the latch rib **182** may be substantially aligned with the container rib **139**. In some examples, the latch rib **182** may also be substantially aligned with the channel rib **160**. Still in other examples, the latch rib **182**, the channel rib **160**, and the container rib **139** may all be substantially aligned with each other to focus the clamping force created by each latch **170** when in its locked configuration onto the gasket **200**. In some such embodiments, the latch rib **182**, the channel rib **160**, and the container rib **139** may be disposed in a common plane when a latch **170** is in its locked configuration.

[0052] FIGS. **8-12** illustrate the container system **100** in an unlocked configuration. When the container system **100** is in an unlocked configuration, one or more latches **170** may be disengaged (e.g., free of contact) from the container **110**. When one or more of the latches **170** are disengaged from the container **110**, the pressure on the gasket **200** is reduced, thus permitting the seal between the gasket **200** and the container **110** to be released. To move the latch **170** to an unlocked configuration from a locked configuration, a user may rotate the latch **170** about its corresponding axles **161** in an upward direction such that the flange **176** is free of contact with the perimeter lip **130** and the latch rib **182** is also free of contact with an outward facing surface **155** opposite the upper surface **158** of the channel **152**. In addition, the container system **100** may also include a restraint to prevent the latches **170** from accidentally moving when they are in an unlocked configuration. Each latch **170** may include a curved wall **184** connected to the top wall **172** opposite the side wall **174**, where the curved wall **184** may have a restraint wall **186** that extends from an outer surface **188** of the curved wall **184**. The restraint wall **186** may contact and in some cases, engage and rides over and/or move beyond a protrusion **169** extending upward from an upper surface **168** of the receiver **148** as the latch **170** is rotated upward. Once the restraint wall **186** contacts or moves beyond the protrusion **169**, the latch **170** may be maintained in an unlocked configuration until a user applies a force to rotate the latch **170** downward toward the locked configuration to overcome the frictional and interference interaction between the restraint wall **186** and the protrusion **169**. In some embodiments, the force applied to rotate a latch **170** toward the locked configuration may be greater than a non-zero threshold force.

[0053] FIGS. **19-22** illustrate an exemplary latch **170**. As shown, the latch **170** may include a top wall **172** connected to a side wall **174**, where the side wall **174** is substantially perpendicular to the top wall **172**. The side wall **174** may further include a flange **176** that may extend inward and away from an inner surface **177** of the side wall **174**. The latch **170** may have a latch length extending from a first end **183** of the top wall **172** to a second end **185** of the top wall **172**. The latch rib **182** may have a latch rib length (e.g., a distance in the same direction as the latch length from the first end **183** to the second end **185**) that is approximately 95 percent of the latch length to exert a consistent force onto the gasket **200**. In some examples, the latch rib length may be within a range of 85 percent and 98 percent of the latch length. The latch rib **182** may also add localized stiffness to the top wall **172** to better allow consistent pressure on the gasket **200** to help maintain a seal. In addition, the latch rib **182** may be centered on the latch length, such that the latch rib **182** is centered upon a plane located at the midpoint of the latch length. The flange **176** may also be centered upon a plane at the midpoint of the latch length as well. The curved wall **184** of each latch **170** may extend the entire latch length as well and connect a portion of each axle clip **178**. The outer surface **188** may include a restraint wall **186** that extends outward, where the restraint wall **186** may also be centered upon the latch length as well. As shown in FIG. **22**, the latch **170** may include a recess **181** that is located adjacent to and inboard of each axle clip **178**. For instance, each latch **170** may include two recesses **181** where each recess **181** is located adjacent a corresponding axle clip **178** in a direction toward a center of the latch **170**. A portion of each recess **181** may be located on an inner surface **180** of the top wall **172** and have a curved surface to allow for the rotation of the latch **170** around end cap **166** of the axle **161**. The axle clips **178** may have a curved inner surface and be configured to pivotally connected to the cylindrical portion **162** of the axle **161** of the lid **142**.

[0054] FIGS. **23-24** illustrate the container **110**. As discussed above, the container **110** may include a container body **111** that has a first wall **112**, a second wall **114** opposite the first wall **112**, a third wall **116** extending between the first wall **112** and the second wall **114**, and a fourth wall **118** opposite the third wall **116** extending between the first wall **112** and the second wall **114**. A bottom wall **120** may connect the first wall **112**, the second wall **114**, the third wall **116**, and the fourth wall **118** at a bottom end **122** of the container **110** forming a storage cavity **124** with a container opening **126** opposite the bottom wall **120** at an upper end **128** of the container **110**. A step **129** may extend along the walls **112**, **114**, **116**, and **118** such that the step is substantially parallel with the bottom wall **120**. The step **129** reduces the width across the container **110** and may allow additional space for a user to fit their hand under the side wall **174** of the latch **170** to easily rotate the latch **170** from a locked configuration to an unlocked configuration. The step **129** may also assist in allowing the multiple containers **110** to stack or nested together for storage.

[0055] In some examples, the number of receiver **148** and latches **170** may be equal to the number of walls of the container **110**. For example, in the illustrated examples shown in FIGS. **1-24**, the lid assembly **140** includes four latches **170** which is equal to the number walls of the container **110**. The container system **100** may have different sizes as shown in FIGS. **25** and **26**. The container systems **300** and **400** illustrated in FIGS. **25** and **26** respectively may have all of the same features as the container system **100** shown in FIGS. **1-24**. As the container system **400** of FIG. **26** is smaller in size than container system **300**, the lid assembly **140** may comprise only two latches and still provide the necessary force to compress the gasket to adequately seal the container system **400**. Optionally, on smaller container systems such as container system **400** illustrated in FIG. **26**, the lid assembly **140** may comprise only two latches **170** that are arranged opposite each other with the smaller sides of the container system **400** not having a latch.

[0056] The container **110**, the lid **142**, and the latches **170** may be formed from a polymeric material using a molding process, such as injection molding or other molding process known to one skilled in the art. The polymeric material may be a clear polymeric material that allows a user to see the contents in the storage cavity **124**. In addition, the polymeric material of the container **110** and the lid **142** may be a food safe material such as Tritan™. In some examples, the container **110** may be formed from a glass material.

[0057] While various embodiments have been described, it will be apparent to those of ordinary skill in the art that many more embodiments and implementations are possible that are within the scope of the claims. The various dimensions described above are merely exemplary and may be changed as necessary. Accordingly, it will be apparent to those of ordinary skill in the art that many more embodiments and implementations are possible that are within the scope of the claims. Therefore, the embodiments described are only provided to aid in understanding the claims and do not limit the scope of the claims.

Claims

1. A container system, comprising: a container comprising: a first wall; a second wall opposite the first wall; a third wall extending between the first wall and the second wall; a fourth wall opposite the third wall extending between the first wall and the second wall; a bottom wall connected to the first wall, the second wall, the third wall, and the fourth wall at a bottom end of the container forming a storage cavity with a container opening opposite the bottom wall; and a perimeter lip extending laterally outward around at least a portion of a perimeter of the container opening, wherein the perimeter lip includes a lip protrusion extending from a lower surface of the perimeter lip; and a lid assembly releasably connected to the container, wherein the lid assembly covers the container opening when connected to the container, wherein the lid assembly comprises: a lid that includes a top surface and a channel on a bottom side of the lid; a gasket received in the channel; and a latch pivotally connected to the lid, wherein the latch includes a latch rib, such that when the latch is engaged with the perimeter lip of the container, the latch rib is located over the channel of the lid.

2. The container system of claim 1, wherein the channel includes an upper surface that has a channel rib that extends away from the upper surface toward the bottom side of the lid, and wherein the latch rib and the channel rib are substantially aligned when the latch is engaged with the lip protrusion of the container.
3. The container system of claim 2, wherein the container includes a container rib that extends upward and away from an upper surface of the perimeter lip.
4. The container system of claim 3, wherein the container rib is substantially aligned with the channel rib and the latch rib.
5. The container system of claim 2, wherein the channel forms a continuous loop, and the channel rib also forms a continuous loop.
6. The container system of claim 3, wherein the container rib and the channel rib contact the gasket on opposite sides of the gasket.
7. The container system of claim 1, wherein the latch rib contacts an outward facing surface of the lid opposite the channel.
8. The container system of claim 1, wherein the lid comprises a receiver that includes a first axle that extends from a first side surface of the receiver and a second axle that extends from a second side surface of the receiver.
9. The container system of claim 8, wherein the first axle includes a cylindrical portion and an end cap at a free end, wherein an axle clip of the latch connects to the cylindrical portion, and wherein the end cap has a maximum width that is larger than a diameter of the cylindrical portion.
10. The container system of claim 1, wherein the latch includes a top wall connected to a side wall, wherein the side wall is substantially perpendicular to the top wall, wherein the side wall includes a flange that engages the perimeter lip, and wherein the latch rib extends outward from an inner surface of the top wall.
11. The container system of claim 10, wherein the latch has a latch length extending from a first end of the top wall to a second end of the top wall, and the latch rib has a latch rib length, wherein the latch rib length is within a range of 85 percent to 98 percent of the latch length.
12. A lid assembly for a covering an opening of a container, the lid assembly comprising: a lid that includes a top surface and a channel on a bottom side of the lid, wherein the channel includes an upper surface that has a channel rib that extends away from the upper surface toward the bottom side of the lid; a gasket received in the channel; and a latch pivotally connected to the lid, wherein the latch is configured to rotate between a locked configuration and an unlocked configuration, wherein the latch includes a latch rib, such that when the latch is in the locked configuration the latch rib contacts an outward facing surface of the lid opposite the channel, and when the latch is in the unlocked configuration, the latch rib is free of contact with the outward facing surface of the lid opposite the channel.
13. The lid assembly of claim 12, wherein the latch rib and the channel rib are substantially aligned when the latch is in the locked configuration.
14. The lid assembly of claim 12, wherein the latch includes a top wall connected to a side wall, wherein the side wall is substantially perpendicular to the top wall, wherein the side wall includes a flange that is configured to engages a perimeter lip of the container when the latch is in the locked configuration.
15. The lid assembly of claim 13, wherein the latch rib is configured to apply a force to the gasket through the channel rib when the latch is moved to the locked configuration.
16. A container system, comprising: a container comprising: a container body including a storage cavity that is accessed through a container opening; and a perimeter lip extending laterally outward around at least a portion of a perimeter of the container opening, wherein the perimeter lip includes: a lip protrusion extending from a lower surface of the perimeter lip; and a container rib that extends upward and away from an upper surface of the perimeter lip; and a lid assembly releasably connected to the container, wherein the lid assembly covers the container opening when

connected to the container, wherein the lid assembly comprises: a lid that includes a top surface, a channel located on a bottom side of the lid, wherein the channel includes an upper surface that has a channel rib that extends away from the upper surface toward the bottom side of the lid, and a plurality of receivers positioned laterally outward of the top surface, wherein a first receiver of the plurality of receivers includes a first side surface and a second side surface, and wherein a first axle extends from the first side surface and a second axle extends from the second side surface; a gasket received in the channel; and a plurality of latches, wherein a first latch of the plurality of latches includes a top wall, a side wall connected to the top wall, and a pair of axle clips extending from an inner surface of the top wall, wherein the side wall includes a flange that engages a bottom surface of the lip protrusion of the perimeter lip when container system is in a locked configuration to compress the gasket between the container rib and the channel rib.

17. The container system of claim 16, wherein the first axle includes a cylindrical portion and an end cap at a free end, wherein an axle clip of the pair of axle clips connects to the cylindrical portion.

18. The container system of claim 16, wherein the first latch further includes a curved wall connected to the top wall opposite the side wall, wherein the curved wall has an outer surface that includes a restraint wall, wherein when the first latch is rotated upward to an unlocked configuration, the restraint wall contacts a protrusion extending upward from an upward facing surface of the first receiver.

19. The container system of claim 16, wherein the first latch further includes a latch rib that extends from the inner surface of the top wall, and wherein when the first latch is in the locked configuration, the latch rib contacts an outward facing surface of the lid opposite the channel.

20. The container system of claim 16, wherein the channel is located laterally outward of the first axle and the second axle of each receiver.